

SECOND ORDER

AI, Capital & Work · A Second Order research briefing

ISSUE 02 · JUNE 12, 2026

The Wrong Denominator

Since Issue 01 we have held that the binding constraint on AI is cost and diffusion, not capability. This week we take that to the unit itself: a token is priced by the seller, not by the value it delivers to the buyer.

Tokens measure lab revenue, not the value buyers pay for

THE THROUGH-LINE

Since Issue 01 we have read AI through cost and diffusion. Token consumption repriced the semiconductor complex and underpins a near-trillion-dollar IPO runway, yet priced per completed task the cost of intelligence is collapsing. Both are true; the gap between them is the story.

WHY NOW

Flat-rate pricing ended June 1 (GitHub) and Anthropic filed its S-1 the same morning; the token spend index rolled over after doubling, and on June 10 Apollo's co-president called token talk "a lot of BS."

WHAT YOU'LL LEARN

Why the bills rise as the unit cheapens, what enterprises cut instead of AI, what the buildout does to the GDP arithmetic (we ran it), and where the dollars pool when good enough will do.

Takeaway: govern cost per completed task, not adoption. Token budgets have begun to behave like headcount budgets: capped, forecast, and fought over.

Both sides of the standoff are right, in different units

“I think tokenmaxxing and token talk is - it’s a lot of BS, honestly. ... Prices are collapsing per unit of IQ, if you did it that way.”

John Zito, co-president, Apollo · Morgan Stanley US Financials Conference, June 10, 2026

Goldman's desk, five days earlier, named the metric: useful task completion per watt and per dollar.

\$37.50 → \$0.175

per M tokens for the cheapest GPT-4-level model, in 23 months (Epoch AI; Exhibit 2)

~13x

growth in average business token spend since Jan 2025 (Ramp); a typical agent job burns ~96,000 tokens (SemiAnalysis)

85%

of Q1 2026 GDP growth came from the IT-investment categories (Second Order calculation; Exhibit 5)

Token expenditure is a near-useless productivity metric and a decisive revenue metric; that gap sets the next 18 months.

Seventy days repriced the cost of AI

EXHIBIT 1

Seventy days from tokenmaxxing to tokenpanic

Key repricing events, April 2 to June 11, 2026

Apr 2	●	OpenAI Codex moves to usage-based billing
Apr 27	●	Developer backlash to Copilot repricing: "pay the same, get less"
May 19	●	Google Gemini moves to usage-based billing
May 26	●	Uber burns its full-year 2026 AI budget in four months
May 28	●	Axios "AI sticker shock"; \$500M single-month Claude bill reported*
May 30	●	Amazon removes its internal token leaderboard (KiroRank)
Jun 1	●	GitHub meters every Copilot plan · Anthropic files confidential S-1 · Walmart caps its agent
Jun 2	●	Uber caps agentic coding tools at \$1,500 per engineer per month, per tool
Jun 8	●	Citrini Research: "tokenmaxxing → tokenpanic"
Jun 10	●	Apollo's Zito: price per unit of IQ, "token talk is BS" · Citadel's Tokenomics note
Jun 11	●	Altman: token costs went from non-issue to "huge issue"; Microsoft pitches token-lean MAI models

THE PANIC
WEEKS

Note: *Single-sourced (Axios, anonymous consultant); unconfirmed by Anthropic. Reported dates; leaderboard removal reported late May.

Source: Company announcements; GitHub changelog; Bloomberg; Axios; Business Insider; Citrini Research; Citadel Securities.

SO WHAT

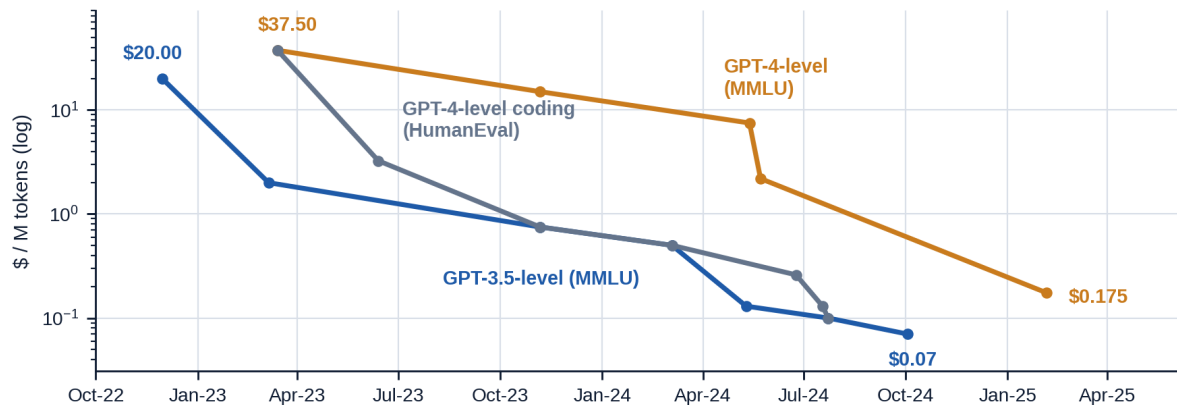
- All four major providers moved to usage billing within 70 days; flat-rate pricing ended June 1.
- Uber capped agentic tools at \$1,500 per engineer per month after burning its 2026 budget in four months.
- Buyers are metering, not retreating: Amazon killed the token leaderboard, Walmart capped its agent, Microsoft cut Claude Code seats.

A fixed capability level deflates 100x or more

EXHIBIT 2

Zito's denominator: a fixed level of capability deflates 100x or more

\$ per million tokens (log) for the cheapest model at or above each capability threshold



Note: Observed model prices only, 3:1 input/output blend; reasoning models excluded by Epoch's method. Series ends Feb 2025 (dataset vintage); no extrapolation drawn. Corroborating anchor (Stanford AI Index): GPT-3.5-class output fell \$20.00 to \$0.07 per M tokens in 18 months, 280x.

Source: Epoch AI / Artificial Analysis (CC-BY); data and vintage notes filed in [data/i02_cost_curve/](#).

SO WHAT

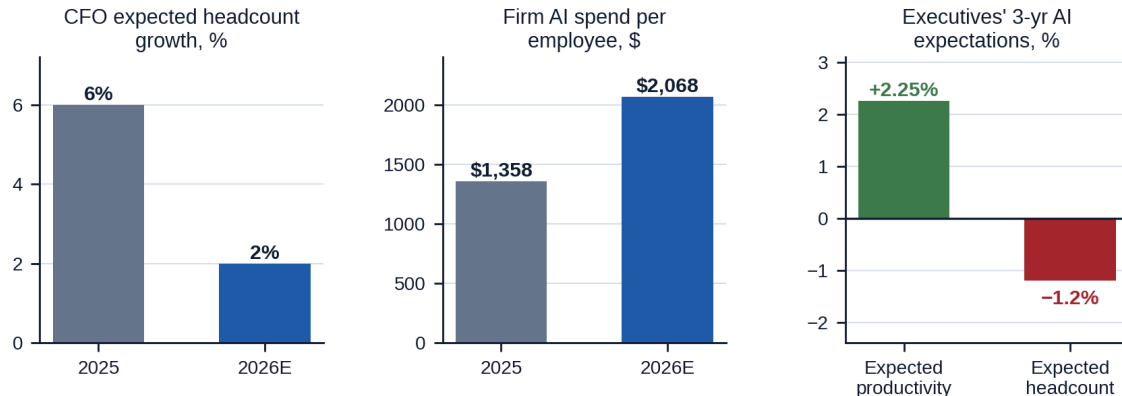
- Cheapest GPT-4-level model: \$37.50 to \$0.175 per M tokens in 23 months (Epoch AI).
- GPT-3.5-level: \$20.00 to \$0.07, the AI Index's 280x in 18 months.
- The steelman holds: Gartner still sees bills rising, since agentic work burns 5-30x the tokens per task and consumption outruns the price decline.

Enterprises ration without retreating: the capital swap

EXHIBIT 3

The capital swap: compute budgets are eating headcount plans

CFO and executive survey readings, 2025 to 2026



Note: Panels mix different surveys, samples, and horizons; read as direction, not level. The \$280B aggregate AI-investment figure in the text is the Atlanta Fed's flagged ballpark estimate.

Source: Gartner CFO surveys (Oct 2025, Feb 2026; n=303); Atlanta Fed Survey of Business Uncertainty macroblog (May 6, 2026); data/i02_surveys/.

SO WHAT

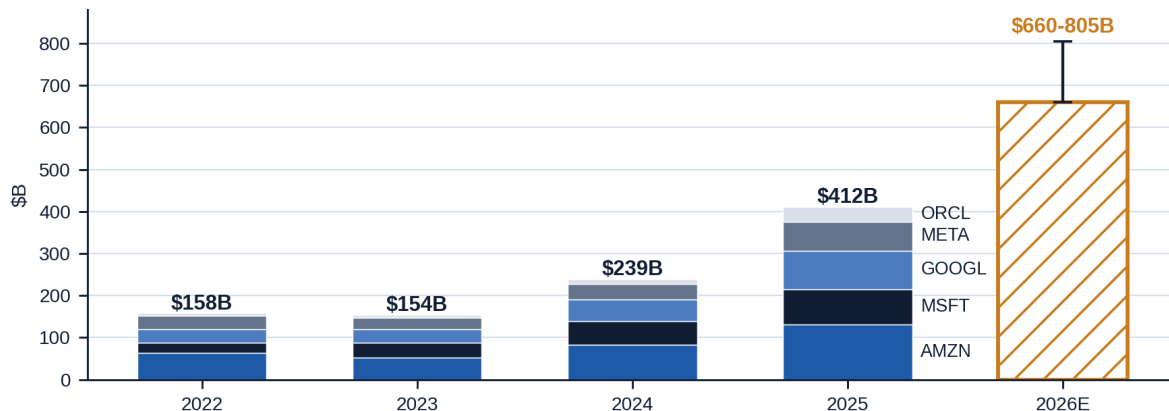
- About 60% of UBS checks call token costs a real issue; none are slamming the brakes (via press coverage; see report caveat).
- They cut around AI: CFO headcount growth expectations collapsed 6% to 2% while 75% raise tech budgets (Gartner).
- Executives expect +2.25% productivity against -1.2% headcount; AI spend per employee rose \$1,358 to \$2,068 (Atlanta Fed).

The buildout nearly tripled in two years

EXHIBIT 4

The denominator debate is funded by a buildout that nearly tripled in two years

Big-5 hyperscaler cash capex (purchases of PP&E), \$B by calendar year, with 2026 guidance range



Note: Cash PP&E only; excludes new finance leases, so levels sit below lease-inclusive compilations (Morgan Stanley's \$805B basis). Oracle fiscal quarters mapped to calendar. 2026E bar = sum of company guidance (~\$660B, estimate); whisker spans to Morgan Stanley's \$805B.

Source: Second Order calculations from SEC EDGAR XBRL filings; 2026 guidance per company earnings calls; data and method in data/i02_capex/.

SO WHAT

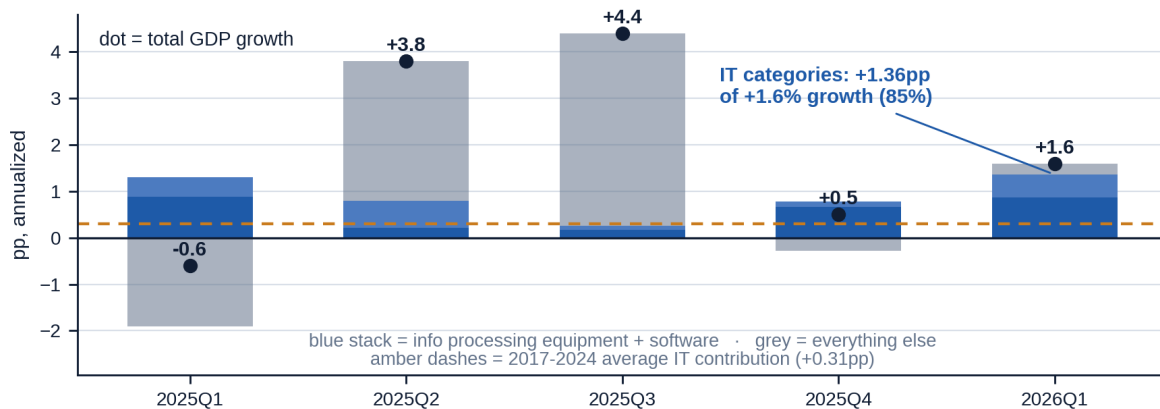
- Big-5 cash capex ran \$158B (2022) to \$412B (2025), computed from SEC filings, not a sell-side deck.
- 2026 company guidance sums to about \$660B; Morgan Stanley's lease-inclusive estimate is \$805B.
- Slok: about 60% of operating cash flow now goes to capex, and the top-10 S&P names trade richer than at the 1999 peak.

We tested the 75% claim ourselves

EXHIBIT 5

We checked the 75% claim: IT investment carried 85% of Q1 growth

Contributions to annualized real GDP growth, percentage points (BEA second estimate, May 28, 2026)



Note: Contributions are gross of imports (Q1 2025: equipment +0.89pp while GDP fell 0.6%, the tariff quarter). Not all IT investment is AI; data-center construction and power sit elsewhere and are excluded. Descriptive, not causal.

Source: Second Order calculations from BEA via FRED (A191RL1Q225SBEA, Y034RY2Q224SBEA, B985RY2Q224SBEA); data/i02_gdp_contrib/.

SO WHAT

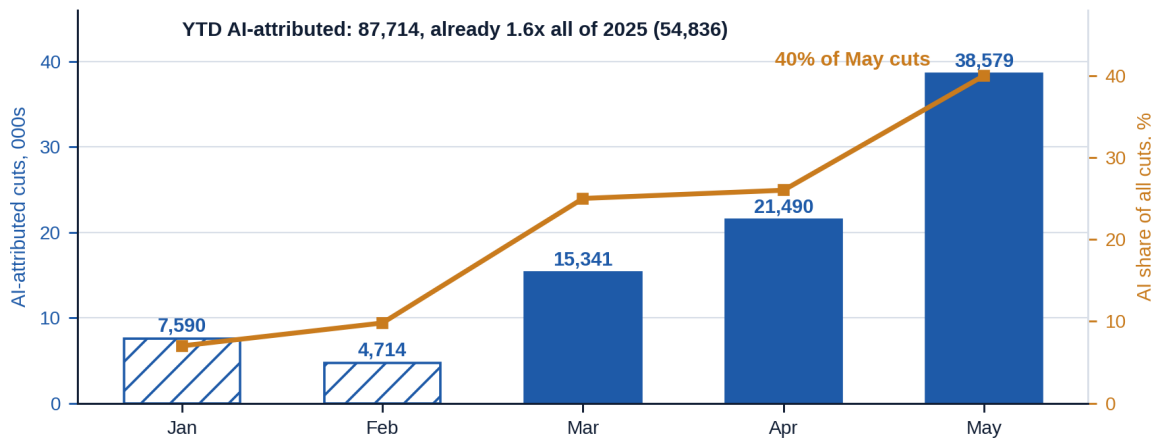
- Info-processing equipment plus software contributed 1.36pp of Q1 2026's 1.6% growth, 85% of it (BEA via FRED).
- The surge over the 2017-2024 trend is about 1.05pp, two thirds of all growth; strip it out and the economy grew about 0.5%.
- Caveats cut both ways: gross of imports, misses data centers and power, includes non-AI IT. Our calc bounds the claim, not settles it.

The narrative is firing ahead of the capability

EXHIBIT 6

The narrative is doing the firing: AI is the No. 1 stated reason for US job cuts

AI-attributed announced job cuts (bars, thousands) and AI share of all cuts (line, %), 2026



Note: Hatched bars are Second Order residuals from Challenger's published year-to-date figures, not reported directly. Attribution is employer self-reporting; Challenger itself flags that AI may cover ordinary cost cuts.

Source: Challenger, Gray & Christmas monthly reports (Mar-May reported); Jan-Feb derived from YTD totals; data/i02_challenger/.

SO WHAT

- AI is the No. 1 stated layoff reason for a third month: 38,579 May cuts, 40% of the total.
- YTD AI-attributed cuts (87,714) already run 1.6x the whole of 2025.
- Self-attribution caveat: firms cut ahead of demonstrated capability because investors reward the story (HBR); some may cite AI to pay for AI.

The operator's playbook for the next two quarters

1 · RE-DENOMINATE

Instrument cost per completed task for your top five AI workflows, and retire the “AI usage” KPIs; Amazon's leaderboard bought token burn, not output.

2 · WRITE A ROUTING POLICY

Default to the cheapest model that clears your quality bar, and escalate to frontier only on documented ROI: a local, mid, frontier ladder.

3 · GOVERN TOKENS LIKE HEADCOUNT

Pair caps, alerts, and dashboards (Uber's \$1,500 template) with workflow redesign, so caps do not freeze adoption at its least productive shape.

4 · RE-UNDERWRITE CONTRACTS

Stress-test anything priced on flat rates at 2-3x usage growth, and track renewal dates: annual Copilot plans roll onto metered billing as they renew.

5 · SET THE LABOR-TO-COMPUTE RATIO

Decide what share of next year's capacity growth is hired versus computed, and own that trade-off deliberately rather than inherit it (Exhibit 3).

The capital swap is happening either way; the only choice is whether you set the ratio or inherit it under bill shock.

WATCHLIST

What would change our mind

- **Silicon Data token index** · already 12% off the peak; a sustained break lower separates deflation in the spend line from demand destruction
- **Challenger June report (early July)** · does AI attribution hold above 40% of cuts, or does the un-attributing begin
- **Q2 earnings (late July)** · hyperscaler capex guidance against softening token expenditure; first quarter the two can diverge
- **BEA Q2 advance estimate (July 30)** · our Exhibit 5 bound updates mechanically; the one-trade economy thesis hardens or cracks
- **Anthropic roadshow** · committed spend versus meterable usage customers are learning to reduce
- **Falsifier** · a sustained break lower in the token index plus a hyperscaler guidance cut would flip our read from deflation to demand destruction

NEXT ISSUE · THE ENTRY-LEVEL PUZZLE Stanford says AI displaces 22-to-25-year-olds (16% relative decline). The NY Fed blames remote work (64% of the rise). Yale sees nothing. Someone is measuring wrong; next week we show you who.